

数学练习 第 03 份 答案与解析

共 68 题 与《练习03》题号一一对应

1. 当 $t \rightarrow +\infty$ 时 $(1+t^2)\sin\frac{1}{t} = t + \frac{5}{6t} + o(\frac{1}{t})$, 故被积函数 $\sim t$, $\int_1^x \sim \frac{x^2}{2}$,

$$\lim_{x \rightarrow \infty} \frac{1}{x^3} \int_1^x \left[(1+t^2)\sin\frac{1}{t} - \cos t \right] dt = 0.$$

(也可用洛必达法则: 极限 $= \lim_{x \rightarrow \infty} \frac{(1+x^2)\sin\frac{1}{x} - \cos x}{3x^2} = 0$.)

2. 由 $\ln y = Cx + x^2$ 得 $C = \frac{\ln y - x^2}{x}$, 又 $\frac{y'}{y} = C + 2x$, 消去 C :

$$\frac{xy'}{y} = \ln y - x^2 + 2x^2 = \ln y + x^2 \implies xy' - y \ln y = x^2 y.$$

选 A.

3. $g = b(y_1 + y_2)^2$ 的秩为 1, 故 $f = (x_1 + 2x_2)^2 + (ax_1 + x_2)^2$ 也必须秩为 1, 即两线性型成比例:

$$\frac{1}{a} = \frac{2}{1} \implies a = \frac{1}{2}.$$

此时 $f = \frac{5}{4}(x_1 + 2x_2)^2$ 半正定, 故必须 $b > 0$. 选 A.

4. 曲线 $x + y + xy = 1$ 即 $y = \frac{1-x}{1+x}$, 与圆弧都过 $(1, 0), (0, 1)$, 直线 $x + y = 1$ 夹在两者之间.

在 $D_1 = D \cap \{x + y \leq 1\}$ 上 (关于 $y = x$ 对称, $\iint (x - y) = 0$):

$$\iint_{D_1} (x - y + 1) d\sigma = S(D_1) = \int_0^1 \left[(1-x) - \frac{1-x}{1+x} \right] dx = \frac{3}{2} - 2 \ln 2.$$

在 $D_2 = D \cap \{x + y > 1\}$ 上用极坐标, 被积函数 $= (\cos \theta + \sin \theta)^2 = c^2$, $\frac{1}{c} \leq r \leq 1$:

$$\iint_{D_2} = \int_0^{\frac{\pi}{2}} c^2 \cdot \frac{1}{2} \left(1 - \frac{1}{c^2} \right) d\theta = \frac{1}{2} \int_0^{\frac{\pi}{2}} \sin 2\theta d\theta = \frac{1}{2}.$$

故 $I = \frac{3}{2} - 2 \ln 2 + \frac{1}{2} = 2 - 2 \ln 2$.

5. $(1+x)^{2x} = e^{2x \ln(1+x)} = 1 + 2x^2 + o(x^2)$, 故

$$\frac{xf(x) - (1+x)^{2x} + 1}{x^2} = \frac{xf(x) - 2x^2 + o(x^2)}{x^2} \rightarrow 1 \implies \lim_{x \rightarrow 0} \frac{f(x)}{x} = 3.$$

由连续性 $f(0) = \lim_{x \rightarrow 0} f(x) = 0$, 于是

$$f'(0) = \lim_{x \rightarrow 0} \frac{f(x) - f(0)}{x} = 3,$$

即 f 在 $x = 0$ 可导且 $f'(0) = 3$.

6. $\{u_n\}$ 正、单调减但未必趋于 0 : A 需 $u_n \rightarrow 0$; B 取 $u_n \equiv 1$ 发散 ; C 取 $u_n = 2^{-n}$ 时通项恒为 $\frac{1}{2}$ 发散 .
D : 由 $u_n - u_{n+1} = (\sqrt{u_n} - \sqrt{u_{n+1}})(\sqrt{u_n} + \sqrt{u_{n+1}}) \leq 2\sqrt{u_n}(\sqrt{u_n} - \sqrt{u_{n+1}})$ 得

$$\frac{u_n - u_{n+1}}{\sqrt{u_n}} \leq 2(\sqrt{u_n} - \sqrt{u_{n+1}}),$$

右端级数 (裂项) 收敛, 故 D 收敛. 选 D.

7. $P\{X = x\} = F(x) - F(x-0)$, 故对一切 x 有 $P\{X = x\} = 0$ 当且仅当 F 处处连续. 选 B.

8. 两式对 x 求导 :

$$2x + 2yy' + 2zz' = 3, \quad 2 - 3y' + 5z' = 0.$$

解得

$$\frac{dz}{dx} = \frac{9 - 6x - 4y}{2(5y + 3z)}, \quad \frac{dy}{dx} = \frac{15 - 10x + 4z}{2(5y + 3z)}.$$

9. 由高斯公式, $\operatorname{div} \gamma = 3$,

$$\iint_S \gamma \cdot \mathbf{n} dS = 3 \cdot \frac{4}{3}\pi = 4\pi.$$

10. (I) $E - A = \begin{pmatrix} 4-a & 1 & -2 \\ 1 & 4-a & -2 \\ 1 & 1 & 1-a \end{pmatrix}$, 秩为 1 要求 $(4-a)^2 = 1$, 即 $a = 3$ 或 5 ;

$a = 5$ 时前两行不成比例, 故 $a = 3$ (此时 $E - A$ 三行全为 $(1, 1, -2)$).

(II) $a = 3$ 时 $A - E$ 的三行均为 $-(1, 1, -2)$, 故 $(A - E)\alpha = -(x_1 + x_2 - 2x_3)(1, 1, 1)^T$.

又 $(A^2 - E)\alpha = (A + E)(A - E)\alpha = (A + E)\beta = 2\beta$ 即 $(A - E)\beta = 0$,

而 $\beta = k(1, 1, 1)^T$ 自动满足. 故

$$\alpha = (x_1, x_2, x_3)^T \text{ 为任一满足 } t = x_1 + x_2 - 2x_3 \neq 0 \text{ 的向量, } \beta = -t(1, 1, 1)^T.$$

11. $\left(\frac{F^2}{2}\right)' = Ff = \frac{\ln \tan x}{\sin x \cos x}$, 注意 $d(\ln \tan x) = \frac{dx}{\sin x \cos x}$, 故

$$\frac{F^2}{2} = \frac{(\ln \tan x)^2}{2} + C; \quad F\left(\frac{\pi}{4}\right) = 0 \Rightarrow C = 0.$$

在 $(\frac{\pi}{4}, \frac{\pi}{2})$ 上 $F > 0$, $\ln \tan x > 0$, 故 $F = \ln \tan x$,

$$f(x) = F'(x) = \frac{1}{\sin x \cos x} = \frac{2}{\sin 2x}.$$

12. 驻点: $z'_x = 3x^2 - 6x = 0$, $z'_y = -6y = 0 \Rightarrow (0, 0), (2, 0)$, z 值为 $0, -4$.

边界 $x^2 + y^2 = 16$ 上 $z = x^3 - 3x^2 - 3(16 - x^2) = x^3 - 48$ ($-4 \leq x \leq 4$), 最大值在 $x = 4$ 处为 16 .

故最大值 $z_{\max} = 16$ (在点 $(4, 0)$ 处取得).

13. 记 $u = \frac{y}{x}$, $z = (x+y)f(u)$, 直接计算得

$$x \frac{\partial^2 z}{\partial x \partial y} + 2y \frac{\partial^2 z}{\partial y^2} = \frac{2y}{x} f'(u) + \frac{y(x+y)}{x^2} f''(u) = \frac{y}{x},$$

即 $(1+u)f''(u) + 2f'(u) = 1$. 令 $p = f'$, $[(1+u)^2 p]' = 1+u$,

$$p = \frac{1}{2} + \frac{C}{(1+u)^2}, \quad f(u) = \frac{u}{2} - \frac{C}{1+u} + D.$$

由 $z(x, x) = 2xf(1) = x$ 得 $f(1) = \frac{1}{2}$; 由 $z'_x|_{(x,x)} = f(1) - 2f'(1) = -\frac{3}{2}$ 得 $f'(1) = 1$, 从而 $C = 2$, $D = 1$:

$$f(u) = \frac{u}{2} - \frac{2}{1+u} + 1.$$

14. 由 $A^n = E$ 知 A 可逆且 $|A|^n = 1$. 因 $A^* = |A|A^{-1}$,

$$(A^*)^n = |A|^n (A^{-1})^n = (A^n)^{-1} = E.$$

15. $\rho = 0$ 故 X, Y 独立, $DX = DY = 2$.

$$\text{Cov}(U, V) = DX - 4DY = 2 - 8 = -6, \quad DU = DV = DX + 4DY = 10,$$

$$\rho_{UV} = \frac{-6}{10} = -\frac{3}{5}.$$

16. $\sqrt[n]{|a_n|} \rightarrow \frac{1}{3}$, 收敛半径 $R = 3$, 收敛区间 $(-3, 3)$.

$x = 3$: 通项 $\frac{1}{n[1+(-\frac{2}{3})^n]} \sim \frac{1}{n} > 0$, 发散;

$x = -3$: 通项 $\frac{(-1)^n}{n[1+(-\frac{2}{3})^n]} = \frac{(-1)^n}{n} + O\left(\frac{(2/3)^n}{n}\right)$, 两部分都收敛, 故收敛 (条件收敛).

即收敛域为 $[-3, 3)$.

17. 设 $A = \iint_D f d\sigma$, 则 $A = \iint_D xy d\sigma + A \cdot S_D$, $S_D = 2$, 得 $A = -\iint_D xy d\sigma$.

而 $\iint_D xy d\sigma = \int_{-1}^1 x \cdot \frac{1-x^2}{2} dx = 0$, 故 $A = 0$, $f(x, y) = xy$.

此时 $\iint_D f(y, x) d\sigma = \iint_D xy d\sigma = 0 = \iint_D f d\sigma$, 而 $\iint_{D_1} f d\sigma > 0$, 故 A、B、D 均不成立. 选 C.

18. 记 $L = b - a$.

$$(I) F_{Z_1} = F^2 \Rightarrow f_{Z_1}(z) = \frac{2(z-a)}{L^2}; \quad 1 - F_{Z_2} = (1 - F)^2 \Rightarrow f_{Z_2}(z) = \frac{2(b-z)}{L^2} \quad (a \leq z \leq b).$$

(II)

$$f(z_1, z_2) = \begin{cases} \frac{2}{(b-a)^2}, & a \leq z_2 < z_1 \leq b, \\ 0, & \text{其他.} \end{cases}$$

19. 由 $f'' > 0$ 知 $f(x) > f(0) + f'(0)x$ ($x \neq 0$), 积分得

$$2 = \int_{-1}^1 f(x) dx > 2f(0) \Rightarrow f(0) < 1.$$

反之任何 $f(0) < 1$ 都可实现. 选 C.

20. $E(X+Y) = 0$,

$$D(X+Y) = DX + DY + 2\rho_{XY}\sqrt{DX DY} = 1 + 4 + 2\left(-\frac{1}{2}\right)(2) = 3,$$

$$P\{|X+Y| \geq 6\} \leq \frac{3}{6^2} = \frac{1}{12}.$$

21. 用极坐标: $r \leq 2 \cos \theta$ ($|\theta| \leq \frac{\pi}{2}$), 高 $h = 1 - \frac{r}{2 \cos \theta}$.

$$V = \int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} \int_0^{2 \cos \theta} \left(1 - \frac{r}{2 \cos \theta}\right) r dr d\theta = \int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} \frac{2}{3} \cos^2 \theta d\theta = \frac{\pi}{3},$$

$$\iiint_{\Omega} x dV = \int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} \frac{2}{3} \cos^4 \theta d\theta = \frac{2}{3} \cdot \frac{3\pi}{8} = \frac{\pi}{4},$$

故 $\bar{x} = \frac{\pi/4}{\pi/3} = \frac{3}{4}$.

22. $\ln \frac{\arctan(x+1)}{\arctan x} = \ln \left(1 + \frac{\arctan(x+1) - \arctan x}{\arctan x}\right) \sim \frac{1/x^2}{\pi/2} = \frac{2}{\pi x^2},$

故 $x^a \cdot \frac{2}{\pi x^2}$ 有非零极限要求 $a = 2$, 此时 $b = \frac{2}{\pi}$. 选 A.

23. 法向量 $\mathbf{n} = (aF'_1, aF'_2, -bF'_1 - cF'_2)$, 而 $(b, c, a) \cdot \mathbf{n} = abF'_1 + acF'_2 - abF'_1 - acF'_2 = 0$, 故切平面恒平行于方向 (b, c, a) , 即直线 $\frac{x}{b} = \frac{y}{c} = \frac{z}{a}$. 选 B.

24. 所给矩阵是特征值 1 的三阶若尔当块, $r(J - E) = 2$.

四个选项中只有 A 满足 $r(M - E) = 2$ (B、C、D 均为 1), 故与之相似的是 A.

25. 直线过 $P_0(0, 0, -1)$, 方向 $s = (3, 3, 2)$, $\overrightarrow{P_0 P} = (1, -1, 1)$,

$$\overrightarrow{P_0 P} \times s = (-5, 1, 6), \quad d = \frac{\sqrt{25+1+36}}{\sqrt{9+9+4}} = \sqrt{\frac{62}{22}} = \frac{\sqrt{341}}{11}.$$

26. $f'(1) = n + n^2$, 切线 $y - (1 + n^2) = (n + n^2)(x - 1)$ 与 x 轴交点

$$a_n = 1 - \frac{1 + n^2}{n(n+1)} = \frac{n-1}{n(n+1)} = -\frac{1}{n} + \frac{2}{n+1}.$$

$\sum a_n x^{n+1}$ 的收敛半径为 1; $x = 1$ 时通项 $\sim \frac{1}{n}$ 发散, $x = -1$ 时为交错级数收敛, 故收敛域 $[-1, 1)$.

$$S(x) = -x \sum_{n \geq 1} \frac{x^n}{n} + 2 \sum_{n \geq 1} \frac{x^{n+1}}{n+1} = x \ln(1-x) + 2[-\ln(1-x) - x] = (x-2) \ln(1-x) - 2x.$$

27. A 的特征值为 $3, -1$; B 的特征值也是 $3, -1$ 且互异, 二者都可对角化, 故相似: 存在可逆 P 使 $P^{-1}AP = B$.

A、C 中 $Q^{-1}AQ$ 、 P^TAP 仍为对称矩阵而 B 不对称; D 需 A 正定而 A 有负特征值. 选 B.

28. 切线交 x 轴于 $N(x - \frac{y}{y'}, 0)$. 由 $|OM| = |ON|$ 及 $y' > 0$ 得 $x - \frac{y}{y'} = -\sqrt{x^2 + y^2}$, 即

$$\frac{dx}{dy} = \frac{x + \sqrt{x^2 + y^2}}{y}.$$

令 $x = uy$: $y \frac{du}{dy} = \sqrt{1 + u^2}$, 积分得 $u + \sqrt{1 + u^2} = Cy$, 即 $x + \sqrt{x^2 + y^2} = Cy^2$.

由 $y(0) = 1$ 得 $C = 1$, 化简得 $y^2 = 2x + 1$, 即

$$y = \sqrt{2x + 1} \quad (\text{满足 } y' > 0, y'' < 0).$$

29. 由对称性 $\iint_S x dS = \iint_S y dS = 0$, 且

$$\iint_S |z| dS = \frac{1}{3} \iint_S (|x| + |y| + |z|) dS = \frac{1}{3} \iint_S dS = \frac{1}{3} \cdot 8 \cdot \frac{\sqrt{3}}{2} = \frac{4\sqrt{3}}{3}.$$

$$\text{故 } I = \frac{4\sqrt{3}}{3}.$$

30. $A^T X = 0$ 的解即与 A 的所有列正交的向量; 两方程组同解

$\iff$ 凡与 A 各列正交者必与 b 正交 $\iff b$ 属于 A 的列空间 $\iff AX = b$ 有解. 选 D.

31. A、B、C 均正确. D 错: $A = PBQ$ 时的列空间为 B 的列空间在 P 下的像, 一般会改变.

例如 $B = \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}$, $P = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$, $Q = E$, 则 $A = \begin{pmatrix} 0 & 0 \\ 1 & 0 \end{pmatrix}$, 两者列向量组不等价. 选 D.

32. $y > 0$ 时

$$F_Y(y) = P\{-\sqrt{y} \leq X \leq \sqrt{y}\} = [\Phi(\sqrt{y}) - \frac{1}{2}] + \frac{1 - e^{-2\sqrt{y}}}{2} = \Phi(\sqrt{y}) - \frac{e^{-2\sqrt{y}}}{2},$$

$y \leq 0$ 时 $F_Y(y) = 0$. 求导得

$$f_Y(y) = \frac{\varphi(\sqrt{y}) + e^{-2\sqrt{y}}}{2\sqrt{y}} \quad (y > 0), \quad f_Y(y) = 0 \quad (y \leq 0).$$

33. $y \neq 0$ 时

$$f'_x(0, y) = \lim_{x \rightarrow 0} \frac{\sqrt{|xy|}}{x} = \lim_{x \rightarrow 0} \frac{\sqrt{|y|}}{\sqrt{|x|}} \cdot \operatorname{sgn} x = \infty \text{ 不存在.}$$

又 $x > 0$ 时 $f'_x = \frac{1}{2}\sqrt{|y/x|}$ (A 差一负号), $x < 0$ 时 $f'_x = -\frac{1}{2}\sqrt{|y/x|}$ (B 差一负号). 选 D.

34. σ^2 未知用 t 检验, 左侧检验的拒绝域为

$$\left\{ \frac{\bar{X} - \mu_0}{S/\sqrt{n}} < -t_\alpha(n-1) \right\}.$$

35. $x = 2$ 对应 $t = 1$, 此时 $y = 3$. 又

$$\frac{dy}{dx} = \frac{4 - 2t}{2t} \Big|_{t=1} = 1,$$

而所求极限 $= \lim_{n \rightarrow \infty} \frac{f(2 + \frac{1}{n}) - f(2)}{1/n} = f'(2) = 1$.

36. 由 $r(AB) = 1 < r(B) = 2$ 知 A 不可逆, $|A| = 2a - 6 = 0 \Rightarrow a = 3$, 此时 $r(A) = 2$, 从而 $r(A^*) = 1$; 又 $r(B) = 2 \Rightarrow r(B^*) = 1$. 于是

$$r \begin{pmatrix} A & O \\ O & B^* \end{pmatrix} = r(A) + r(B^*) = 2 + 1 = 3.$$

选 B.

37. $x = \int_0^t 2e^{-u^2} du \sim 2t$, $y = \int_0^t \sin(t-u) du = 1 - \cos t \sim \frac{t^2}{2}$, 故

$$y \sim \frac{1}{2} \left(\frac{x}{2} \right)^2 = \frac{x^2}{8},$$

即 $f(x)$ 与 x^2 同阶但不等价. 选 C.

38. $y = e^{-ax} [C + \int_0^x e^{as} f(s) ds]$, 由洛必达法则

$$\lim_{x \rightarrow +\infty} y = \lim_{x \rightarrow +\infty} \frac{e^{ax} f(x)}{ae^{ax}} = \frac{b}{a},$$

故有水平渐近线 $y = \frac{b}{a}$. 选 C.

39. $|X_i - \mu|$ 服从半正态分布, $E|X_i - \mu| = \sigma \sqrt{\frac{2}{\pi}}$, 故

$$E\hat{\sigma} = kn\sigma \sqrt{\frac{2}{\pi}} = \sigma \Rightarrow k = \frac{1}{n} \sqrt{\frac{\pi}{2}}.$$

40. (I) $R = 2$; $x = 2$ 时 $\sum \frac{(-1)^n}{\sqrt{n}}$ 收敛, $x = -2$ 时 $\sum \frac{1}{\sqrt{n}}$ 发散. 收敛域 $(-2, 2]$.

(II) $R = 3$ (中心 3); $x = 6$ 时 $\sum \frac{1}{n}$ 发散, $x = 0$ 时 $\sum \frac{(-1)^n}{n}$ 收敛. 收敛域 $[0, 6)$.

(III) $\sqrt[n]{n^n} = n \rightarrow \infty$, $R = 0$, 收敛域 $\{0\}$.

(IV) $\sum \frac{x^{2n-1}}{3^n}$ 当 $x^2 < 3$ 收敛, 端点 $x = \pm\sqrt{3}$ 时通项不趋于 0. 收敛域 $(-\sqrt{3}, \sqrt{3})$.

41. 取 $x = R \cos \theta$, $y = R \sin \theta$, θ 由 $\frac{3\pi}{4}$ 变到 0.

$$\begin{aligned} \int_L y ds &= \int_0^{\frac{3\pi}{4}} R \sin \theta \cdot R d\theta = R^2 \left(1 + \frac{\sqrt{2}}{2} \right) = \frac{(2 + \sqrt{2})R^2}{2}; \\ \int_L y dx &= \int_{\frac{3\pi}{4}}^0 R \sin \theta (-R \sin \theta) d\theta = R^2 \int_0^{\frac{3\pi}{4}} \sin^2 \theta d\theta = R^2 \left(\frac{3\pi}{8} + \frac{1}{4} \right) = \frac{(3\pi + 2)R^2}{8}. \end{aligned}$$

42. $X - Y \sim N(0, 1)$, 故

$$D(|X - Y|) = E(X - Y)^2 - (E|X - Y|)^2 = 1 - \left(\sqrt{\frac{2}{\pi}} \right)^2 = 1 - \frac{2}{\pi}.$$

43. (I) 夹逼: $\frac{\sum k}{n^2 + 2n} \leq \sum_{k=1}^n \frac{k}{n^2 + n + k} \leq \frac{\sum k}{n^2 + n + 1}$, 两端均 $\rightarrow \frac{1}{2}$, 故极限 $= \frac{1}{2}$.

$$(II) \sqrt{\frac{n(n+1)}{2}} - \sqrt{\frac{n(n-1)}{2}} = \frac{n}{\sqrt{\frac{n(n+1)}{2}} + \sqrt{\frac{n(n-1)}{2}}} \rightarrow \frac{\sqrt{2}}{2}.$$

$$(III) \sum_{k=1}^n \frac{1}{2} \left(\frac{1}{2k-1} - \frac{1}{2k+1} \right) \rightarrow \frac{1}{2}.$$

(IV) 若为平方根, 因调和级数发散, 极限为 $+\infty$; 若原题为 n 次根号 $\sqrt[n]{H_n}$, 则极限为 1 (题干待核对).

44. 交换积分次序: $0 < y < \frac{\pi}{4}$ 时 $0 < x < \tan y$,

$$I = \int_0^{\frac{\pi}{4}} \csc 2y \left[\int_0^{\tan y} \frac{1}{3} x^{-\frac{2}{3}} dx \right] dy = \int_0^{\frac{\pi}{4}} \frac{(\tan y)^{\frac{1}{3}}}{\sin 2y} dy \stackrel{t=\tan y}{=} \frac{1}{2} \int_0^1 t^{-\frac{2}{3}} dt = \frac{3}{2}.$$

45. 由 $AB = (\alpha_1, \alpha_2 - 2\alpha_3, \alpha_3 - 2\alpha_2)$ 得 $A = BMB^{-1}$, 其中 $M = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & -2 \\ 0 & -2 & 1 \end{pmatrix}$,

故 A 的特征值与 M 相同, 为 $1, 3, -1$, 正、负惯性指数为 $2, 1$. 而 $\text{tr}(AXX^T) = X^TAX$, 故规范形为 $y_1^2 + y_2^2 - y_3^2$. 选 B.

46. 在 (x_0, y_0, z_0) 处切平面 $\frac{xx_0}{a^2} + \frac{yy_0}{b^2} + \frac{zz_0}{c^2} = 1$, 三截距为 $\frac{a^2}{x_0}, \frac{b^2}{y_0}, \frac{c^2}{z_0}$,

$$V = \frac{a^2 b^2 c^2}{6x_0 y_0 z_0}.$$

由均值不等式, 约束下 $x_0 y_0 z_0$ 在 $\frac{x_0^2}{a^2} = \frac{y_0^2}{b^2} = \frac{z_0^2}{c^2} = \frac{1}{3}$ 时最大, $x_0 y_0 z_0 = \frac{abc}{3\sqrt{3}}$, 故

$$V_{\min} = \frac{\sqrt{3}}{2} abc.$$

47. (I) 设 $k_1\beta + k_2A\beta + k_3A^2\beta = 0$, 即 $\sum_{i=1}^3 (k_1 + k_2\lambda_i + k_3\lambda_i^2)\alpha_i = 0$.

因 $\alpha_1, \alpha_2, \alpha_3$ 线性无关, 得三个方程, 其系数行列式为范德蒙德行列式 $\prod_{i < j} (\lambda_j - \lambda_i) \neq 0$, 故 $k_1 = k_2 = k_3 = 0$.

(II) 由 $A^3\beta = A\beta$ 得 $\sum (\lambda_i^3 - \lambda_i)\alpha_i = 0$, 故 $\lambda_i^3 = \lambda_i$, 即 $\lambda_i \in \{0, 1, -1\}$, 又三者互异, 故特征值为 $0, 1, -1$.

$A - E$ 的特征值为 $-1, 0, -2$, 故 $r(A - E) = 2$.

48. $\left| \frac{\sqrt{n+1} - \sqrt{n-1}}{n} \sin(n+k) \right| \leq \frac{2}{n(\sqrt{n+1} + \sqrt{n-1})} \sim \frac{1}{n^{3/2}},$

由比较判别法级数绝对收敛 (与 k 无关). 选 A.

49. 由 $\frac{\partial P}{\partial y} = \frac{\partial Q}{\partial x}$ 可定出 $k = -1$. 此时

$$P = \frac{x}{y\sqrt{x^2+y^2}}, \quad Q = -\frac{x^2}{y^2\sqrt{x^2+y^2}}, \quad u(x, y) = \frac{\sqrt{x^2+y^2}}{y} + C.$$

故

$$I = u(2, 2) - u(1, 1) = \sqrt{2} - \sqrt{2} = 0.$$

50. (I) ①: $x_1 = -x_2, x_4 = x_2, x_3$ 自由, 基础解系 $(-1, 1, 0, 1)^T, (0, 0, 1, 0)^T$;

②: $x_1 = x_2 - x_3, x_4 = x_3 - x_2$, 基础解系 $(1, 1, 0, -1)^T, (-1, 0, 1, 1)^T$.

(II) 联立四个方程得 $x_1 = -x_2, x_4 = x_2, x_3 = 2x_2$, 故非零公共解为

$$k(-1, 1, 2, 1)^T, \quad k \neq 0.$$

51. $P\{|X - Y| \leq b\} = 1 - \frac{(a-b)^2}{a^2}, \quad P\{\min(X, Y) \leq b\} = 1 - \frac{(a-b)^2}{a^2},$

二者相等, 故差为 0.

52. 分离变量 $y dy = -x dx$, 得 $x^2 + y^2 = C$. 选 C.

53. 本题需要原书图 2-1 (f'' 的图形) 才能确定: 拐点个数 = f'' 变号的点的个数 ($f'' = 0$ 但不变号者不算, f'' 不存在但两侧变号且 f 连续者也算).
因原始材料无图, 答案待核对.

54. 先对 x 积分 ($0 \leq x \leq 1 - y - z$):

$$\begin{aligned} I &= \iint_{y, z \geq 0, y+z \leq 1} (1-y)(1-y-z)e^{-(1-y-z)^2} dy dz \stackrel{u=1-y-z}{=} \int_0^1 (1-y) \int_0^{1-y} u e^{-u^2} du dy \\ &= \int_0^1 s \cdot \frac{1-e^{-s^2}}{2} ds = \frac{1}{4} - \frac{1}{4}(1-e^{-1}) = \frac{1}{4e}. \end{aligned}$$

55. $\frac{1}{\sigma^2} [\sum (X_i - \bar{X})^2 + \sum (Y_j - \bar{Y})^2] \sim \chi^2(n_1 + n_2 - 2)$, 故 $T = \frac{\sigma^2}{n_1 + n_2 - 2} \chi^2(n_1 + n_2 - 2)$,

$$D(T) = \frac{\sigma^4}{(n_1 + n_2 - 2)^2} \cdot 2(n_1 + n_2 - 2) = \frac{2\sigma^4}{n_1 + n_2 - 2}.$$

56. 原书图 9-1 缺失, 按常见图形 (L_1 : 上半圆自 $(1, 0)$ 到 $(-1, 0)$; L_3 : 下半圆自 $(-1, 0)$ 到 $(1, 0)$; L_2 : 整圆或上半圆, 第一型积分) 可判断:

由 $f > 0$ 及 $f(x, -y) = f(x, y)$ 得 $I_3 = -I_1 > 0 > I_1$, 又 $ds \geq |dx|$ 故 $I_2 > I_3$,

从而 $I_2 > I_3 > I_1$, 选 B (图形请核对).

57. $B = AM$, 其中 $M = \begin{pmatrix} 1 & 0 & 0 \\ 1 & 1 & 1 \\ 0 & 1 & 2 \end{pmatrix}$, $|M| = 1$. 于是

$$A^{-1} + B^{-1} = (E + M^{-1})A^{-1}, \quad |A^{-1} + B^{-1}| = \frac{|M + E|}{|M|} \cdot \frac{1}{|A|} = |M + E|.$$

$$M + E = \begin{pmatrix} 2 & 0 & 0 \\ 1 & 2 & 1 \\ 0 & 1 & 3 \end{pmatrix}, \quad |M + E| = 10, \text{ 故}$$

$$|(A^{-1} + B^{-1})^*| = |A^{-1} + B^{-1}|^{3-1} = 100.$$

58. $|A| = a - 3$.

(I) 当 $a = 3$ 时 $\alpha_3 = 2\alpha_2 - \alpha_1$, $r(\alpha_1, \alpha_2, \alpha_3) = 2$; 此时 $\beta_1 = -3\alpha_1 + 2\alpha_2$ 总可表示, 而 $\beta_2 = (1, 0, b)^T$ 可表示当且仅当 $b = 1$.

故 $a = 3$ 且 $b \neq 1$ 时 β_1, β_2 不能同时被线性表示.

(II) 当 $a \neq 3$ 时 A 可逆, 均可唯一表示:

$$\beta_1 = -3\alpha_1 + 2\alpha_2, \quad \beta_2 = \left(1 + \frac{b-1}{a-3}\right)\alpha_1 - \frac{2(b-1)}{a-3}\alpha_2 + \frac{b-1}{a-3}\alpha_3;$$

当 $a = 3, b = 1$ 时也可表示 (有无穷多种): $\beta_1 = -3\alpha_1 + 2\alpha_2, \beta_2 = \alpha_1 + t(\alpha_1 - 2\alpha_2 + \alpha_3)$.

59. $f(0, 1) = 0$; $f'_x(0, 1) = \lim_{x \rightarrow 0} \frac{x}{x} = 1$, $f'_y(0, 1) = \lim_{y \rightarrow 1} \frac{(y-1)\arcsin 0}{y-1} = 0$, 故 A 错.

又

$$\frac{|f(x, y) - f(0, 1) - x|}{\rho} = \frac{|y-1| \arcsin \sqrt{|x|/y}}{\rho} \leq \arcsin \sqrt{\frac{|x|}{y}} \rightarrow 0,$$

故 f 在 $(0, 1)$ 可微且 $df|_{(0,1)} = dx$. 选 C.

60. 记 $g(u) = \int_0^u f$ (偶), $h(u) = \int_0^u tf(t)dt$ (奇, 因 $tf(t)$ 为偶函数).
 $\int_a^x h(u)du = H(x) - H(a)$, 其中 $H(x) = \int_0^x h$ 为偶函数, 故该式为偶函数;
 而 $\int_a^x g(u)du$ 、 $\int_0^x [g(u) - g(a)]du$ 均为奇函数 (相差常数).
 故选含 $tf(t)$ 的那一项, 即 A (A 与 D 在转写中相同, 原书应有细节差别, 请核对).

61.
$$P\{X \leq k+1 \mid X > k\} = \frac{e^{-k} - e^{-(k+1)}}{e^{-k}} = 1 - e^{-1}.$$

62. 令 $t = x - 1 : f = \frac{t}{2-t} = \frac{t}{2} \cdot \frac{1}{1-\frac{t}{2}} = \sum_{n=1}^{\infty} \frac{(x-1)^n}{2^n}, |x-1| < 2$.
 故 $f^{(n)}(1) = n! \cdot \frac{1}{2^n} = \frac{n!}{2^n} (n \geq 1)$.

63.
$$\int_{\frac{1}{2}}^{\frac{\sqrt{3}}{2}} \frac{x^2}{\sqrt{1-x^2}} dx = \frac{1}{2} \left[\arcsin x - x\sqrt{1-x^2} \right]_{\frac{1}{2}}^{\frac{\sqrt{3}}{2}} = \frac{1}{2} \left(\frac{\pi}{3} - \frac{\pi}{6} \right) = \frac{\pi}{12},$$

区间长度为 $\frac{\sqrt{3}-1}{2}$, 故平均值

$$\bar{y} = \frac{\pi/12}{(\sqrt{3}-1)/2} = \frac{\pi}{6(\sqrt{3}-1)} = \frac{(\sqrt{3}+1)\pi}{12}.$$

64. (I) 方程两边除以 $x^2 : \left(\frac{f}{x}\right)' = \frac{a(1-\ln x)}{x^2} + 1$, 而 $\int \frac{1-\ln x}{x^2} dx = \frac{\ln x}{x}$, 故

$$\frac{f}{x} = \frac{a \ln x}{x} + x + C \implies f(x) = a \ln x + x^2 + Cx.$$

由 $f(1) = 1 - a$ 得 $C = -a$, 故 $f(x) = x^2 + a(\ln x - x)$.

(II) $f(x) = 0 \iff a = \varphi(x) := \frac{x^2}{x - \ln x}$ (注意 $x - \ln x > 0$).

$$\varphi'(x) = \frac{x(x+1-2\ln x)}{(x - \ln x)^2} > 0,$$

故 φ 在 $(0, +\infty)$ 上严格增, 且 $\varphi(0^+) = 0, \varphi(+\infty) = +\infty$.

于是方程有唯一实根当且仅当 $a > 0$, 即 $a \in (0, +\infty)$.

65. (I) X, Y 独立同服从 $U[0, 2]$,

$$f_Z(z) = \begin{cases} \frac{z}{4}, & 0 \leq z \leq 2, \\ \frac{4-z}{4}, & 2 < z \leq 4, \\ 0, & \text{其他.} \end{cases}$$

(II) $EX = EY = 1, EX^2 = EY^2 = \frac{4}{3},$

$$E(Z^2) = EX^2 + 2EXEY + EY^2 = \frac{4}{3} + 2 + \frac{4}{3} = \frac{14}{3}.$$

66. 当 $m > n$ 时 $r(AB) \leq \min\{r(A), r(B)\} \leq n < m$, 故 m 阶方阵 AB 不可逆, $|AB| = 0$. 选 B.

67. (I) 方向导数在 $l = (0, 1)$ 方向取最大值 2 说明 $\nabla f(2, 1) = (0, 2)$:

$$f'_x(2, 1) = 4 + a = 0, \quad f'_y(2, 1) = 2a + 2b = 2 \implies a = -4, b = 5.$$

(II) 此时 $f = X^T \begin{pmatrix} 1 & -2 \\ -2 & 5 \end{pmatrix} X = 1$, 矩阵特征值 $\lambda = 3 \pm 2\sqrt{2}$,
距离平方 $d^2 \in [\frac{1}{\lambda_{\max}}, \frac{1}{\lambda_{\min}}] = [3 - 2\sqrt{2}, 3 + 2\sqrt{2}]$, 故

$$d_{\min} = \sqrt{2} - 1, \quad d_{\max} = \sqrt{2} + 1.$$

68. D 为上半圆盘 $(x-1)^2 + y^2 \leq 1 (y \geq 0)$, 直线 $x+y=2$ 交其边界于 $(1, 1), (2, 0)$.

$$\iint_D (2-x-y) d\sigma = 2 \cdot \frac{\pi}{2} - 1 \cdot \frac{\pi}{2} - \frac{4}{3\pi} \cdot \frac{\pi}{2} = \frac{\pi}{2} - \frac{2}{3}.$$

记 $D_2 = D \cap \{x+y > 2\}$, 作平移 $u = x-1, v = y$ 并用极坐标 ($\frac{1}{c} \leq r \leq 1, c = \cos\theta + \sin\theta$) :

$$\iint_{D_2} (x+y-2) d\sigma = \int_0^{\frac{\pi}{2}} \left(\frac{c}{3} - \frac{1}{2} + \frac{1}{6c^2} \right) d\theta = \frac{2}{3} - \frac{\pi}{4} + \frac{1}{6} = \frac{5}{6} - \frac{\pi}{4}.$$

故

$$I = \iint_D (2-x-y) d\sigma + 2 \iint_{D_2} (x+y-2) d\sigma = \left(\frac{\pi}{2} - \frac{2}{3} \right) + \left(\frac{5}{3} - \frac{\pi}{2} \right) = 1.$$